

AI for Office Automation

DAY 3 — TASKS WORKBOOK

Session Duration: 4 Hours | Hands-On Practice

This workbook contains 10 structured tasks covering AI-powered data analysis, insight extraction, mini-automation building, no-code tool design, and a Capstone Workflow Lab. Tasks are designed for both general business users and software developers — with dedicated exercises for each track.

Workbook Structure

Section 1	AI + Data Analysis	Tasks 1–3	~60 min
Section 2	Insight Extraction & Reports	Tasks 4–5	~45 min
Section 3	Building Mini-Automations	Tasks 6–7	~60 min
Section 4	No-Code Automation Design	Task 8	~30 min
Section 5	Capstone Workflow Lab	Tasks 9–10	~45 min

■ How to Use This Workbook

Symbol	Meaning
■ Blue box	General business track — relevant to all participants
■■■ Purple box	Developer track — additional exercises for software developers
■ Yellow box	Tip, warning, or important guidance
■ Green box	Sample data provided — use this for the exercise
[blank box]	Space for your written answer or prompt
■ Time	Suggested time allocation for the task
● Difficulty	Foundation / Intermediate / Advanced / Capstone

■ Before You Start

You will need access to an AI tool for all tasks in this workbook. Recommended: **ChatGPT-4o** or **Claude** — both support file uploads. For Tasks 1–5, you will use the sample datasets provided in the workbook. For Tasks 6–10, you will design workflows for your own work context. All tasks can be completed without uploading real company data.

SECTION 1 | AI + Data Analysis

In this section you will practice uploading and interpreting data using AI tools. You will use the sample Sales Dataset provided below for Tasks 1–3.

■ Sample Dataset — Weekly Sales Data (Use for Tasks 1, 2 & 3)

Copy and paste the following CSV data into your AI tool, or type 'use the sales dataset I provided earlier' in follow-up prompts.

order_id	date	region	product	category	qty	unit_price	revenue	rep_name	status
1001	2024-03-01	North	Widget A	Hardware	12	45.00	540.00	sara ali	Closed
1002	2024-03-01	South	Widget B	Software	5	120.00	600.00	AHMED NASSER	Closed
1003	03/02/2024	North	Gadget X	Hardware	8	89.99	719.92	Sara Ali	Closed
1004	2024-03-03	East	Widget A	Hardware	,,	0.00		Khalid Omar	Pending
1005	2024-03-04	West	Service Pack	Services	2	250.00	500.00	nour hassan	Closed
1006	2024-03-05	South	Gadget X	Hardware	15	89.99	1349.85	Ahmed Nasser	Closed
1007	2024-03-05	North	Widget B	Software	3	120.00	360.00	sara ali	Closed
1008	2024-03-06	East	Service Pack	Services	1	250.00	250.00	KHALID OMAR	Closed
1009	2024-03-07	West	Widget A	Hardware	20	45.00	900.00	Nour Hassan	Closed
1010	2024-03-08	North	Gadget X	Hardware	6	89.99	N/A	Sara Ali	pending
1011	2024-03-08	South	Widget B	Software	8	120.00	960.00	Ahmed Nasser	Closed
1012	2024-03-09	East	Widget A	Hardware	4	45.00	180.00	khalid omar	Closed
1013	2024-03-10	West	Gadget X	Hardware	11	89.99	989.89	Nour Hassan	Closed
1014	2024-03-11	North	Service Pack	Services	3	250.00	750.00	Sara Ali	Closed
1015	2024-03-11	South	Widget A	Hardware	7	45.00	315.00	ahmed nasser	Closed
1016	2024-03-12	East	Widget B	Software	2	120.00	240.00	Khalid Omar	Closed
1017	2024-03-13	West	Service Pack	Services	5	250.00	1250.00	Nour Hassan	Closed
1018	2024-03-14	North	Gadget X	Hardware	9	89.99	809.91	Sara Ali	Closed
1019	2024-03-15	South	Widget A	Hardware	99	45.00	4455.00	Ahmed Nasser	Closed
1020	2024-03-15	East	Service Pack	Services	,,	250.00	0.00	Khalid Omar	Pending

■ What's wrong with this data? — Spot the Issues Before You Start

This dataset has been deliberately designed with quality issues that AI should catch. As you work through Tasks 1–3, watch for: inconsistent date formats, mixed-case names, missing values, a revenue value of 'N/A', an outlier quantity, and inconsistent status casing. Part of your job is to verify that AI correctly identifies these.

1

Data Upload & Quality Assessment

■ General + ■■ Developer ● Foundation ■ 20 minutes

OBJECTIVE

Practice uploading a dataset to an AI tool, asking it to describe the structure, and identify data quality issues before any analysis begins.

PART A — Write Your Upload Prompt

Using the CRTFC framework, write a complete prompt that: (1) gives the AI context about the dataset, (2) asks it to describe the structure, (3) asks it to identify all data quality issues it can find.

Write your upload + quality assessment prompt here:

PART A Prompt:
Context: Weekly sales dataset for business analysis.
Role: Data quality analyst.
Task: Describe structure and identify all data issues.
Format: Rows, columns, issues and recommendations.
Constraints: Do not invent data.

PART B — AI Output Analysis

Rows: 20, Columns: 10.
After running your prompt, answer these questions:
Issues: date formats, name casing, status casing, missing qty, N/A revenue, qty outlier 99.

Question	Your Answer
How many rows and columns did AI identify?	
List ALL data quality issues AI found:	
Which issues did AI miss (if any)?	
Did AI correctly describe what the data represents?	

PART C — Developer Extension

■■■ Developer Task — Write a Data Validation Script

Ask the AI to generate a **Python pandas script** that programmatically validates the dataset. The script should: (1) Load the CSV. (2) Check for missing values per column. (3) Detect inconsistent casing in the 'rep_name' and 'status' columns. (4) Identify rows where 'revenue' is non-numeric. (5) Print a validation report.

■■■ Write your Python validation script request prompt:

Developer validation request:

Create a pandas script to load CSV, check missing values, detect casing problems, detect non-numeric revenue, and print a validation report.

After running the prompt, paste the key output or summarize what the script does:

Output summary:

Script output summary:

Script validates columns, missing values, casing consistency, and invalid revenue.

2

Data Cleaning via AI

■ General + ■■ Developer ● Intermediate ■ 20 minutes

OBJECTIVE

Use AI to clean the sales dataset: fix inconsistent casing, standardize date formats, handle missing values, and flag the outlier. Then compare the cleaned result.

PART A — The Cleaning Prompt Chain

Write a **two-turn cleaning workflow**. Turn 1 asks AI to list all cleaning actions it will take. Turn 2 instructs AI to perform all cleaning and output the cleaned CSV.

Turn 1 — Cleaning Plan prompt:

Turn 1: List cleaning actions before changes.
Turn 2: Apply cleaning and output cleaned CSV.

Verification:

rep_name: Title Case ?

status: Closed / Pending

dates: YYYY-MM-DD ?

missing values: flagged ?

N/A revenue: flagged ?

qty 99: outlier flagged ?

Decision: Keep order 1019 but flag it because it may be a real large order.

PART B — Cleaning Verification

After cleaning, verify the AI's work by answering these questions:

Cleaning Item	Expected Result	AI Result ✓/X	Notes
rep_name casing	All Title Case		
status casing	All 'Closed' / 'Pending'		
Date formats	All YYYY-MM-DD		
Missing qty/revenue	Flagged or filled		
Revenue 'N/A'	Converted or flagged		
Outlier qty (99)	Flagged as outlier		

■ Reflection — The Outlier Decision

Order 1019 has qty = 99 — nearly 5x the next highest order. AI should flag this, but should it remove it? In the box below, write your decision and reasoning:

Your outlier decision and reasoning:

PART C — Developer Extension

■■■ Write a Pandas Cleaning Script

Ask AI to generate a complete Python pandas cleaning script for this dataset that: (1) Applies Title Case to rep_name and Proper Case to status. (2) Parses all dates to YYYY-MM-DD. (3) Converts revenue 'N/A' to NaN. (4) Fills missing qty/revenue with 0 and flags the row with a new column 'data_issue=True'. (5) Flags rows where qty > 50 as potential outliers. (6) Saves cleaned output to sales_clean.csv.

■■■ Developer cleaning script prompt:

3

Data Analysis & Insight Extraction

■ General + ■■ Developer ● Intermediate ■ 20 minutes

OBJECTIVE

Using the (cleaned) sales dataset, apply the 6-step Insight Extraction workflow to produce a ranked analysis and 5 actionable business insights.

PART A — Run the 6-Step Workflow

Execute each step as a separate prompt turn. Record the key output of each step.

Step	Your Prompt (summarized)	Key Output from AI
1 — Describe		
2 — Summarize		
3 — Rank		
4 — Trend		
5 — Anomaly		
6 — Insight		

PART B — Your Top 5 Insights

Based on AI's analysis, write your final 5 insights in the table below. Use your own words — do not just copy the AI output:

#	Insight (your words)	Supporting data point	Recommended action
1			

2			
3			
4			
5			

PART C — Developer Extension

■■■ Generate a Python Analysis Script

Ask AI to write a pandas + matplotlib script that produces: (1) A bar chart of total revenue by region. (2) A line chart of daily revenue over the date range. (3) A summary statistics table exported to analysis_report.xlsx. The script should run end-to-end without errors on the cleaned CSV.

■■■ Analysis script request prompt:

SECTION 2 | Insight Extraction & Report Generation

4

Customer Feedback Analysis

■ General + ■■ Developer ● Intermediate ■ 25 minutes

OBJECTIVE

Analyze the customer feedback dataset using AI: categorize entries, assess sentiment, identify top issues, and produce a structured management report.

■ Sample Dataset — Customer Feedback (15 entries)

Copy and paste this into your AI tool:

```
ID,Source,Date,Feedback Text
F001,Email,2024-03-01,"The onboarding process was smooth and the team was very helpful. Loved the welcome kit!"
F002,Survey,2024-03-02,"Delivery took 3 weeks instead of the promised 5 days. Very disappointed."
F003,Survey,2024-03-03,"Product quality is excellent but the packaging was damaged when it arrived."
F004,Support,2024-03-04,"I have been waiting 2 weeks for a refund. Nobody is responding to my emails."
F005,Email,2024-03-05,"Great experience overall. The mobile app is intuitive and easy to navigate."
F006,Survey,2024-03-06,"Prices are competitive but I wish there were more payment options available."
F007,Support,2024-03-07,"The product stopped working after 10 days. Replacement process is complicated."
F008,Email,2024-03-08,"Customer service representative was rude and unhelpful when I called."
F009,Survey,2024-03-09,"Fast shipping and the product is exactly as described. Will definitely reorder."
F010,Support,2024-03-10,"Cannot log into my account. Reset password link does not work. Urgent please."
F011,Email,2024-03-11,"The new product update is amazing. Performance has improved significantly."
F012,Survey,2024-03-12,"Instructions manual is confusing. Took 2 hours to set up a simple device."
F013,Support,2024-03-13,"Website keeps crashing on mobile. I cannot complete my purchase."
F014,Email,2024-03-14,"Best customer support I have ever experienced. Issue resolved in 10 minutes!"
F015,Survey,2024-03-15,"Good product but overpriced compared to competitors offering similar features."
```

PART A — Categorization & Sentiment Prompt

Write a prompt that asks AI to produce the per-entry categorization table (ID | Category | Sentiment | Key Theme | Priority) from the Feedback Categorization Template.

Your categorization prompt:

PART B — Record AI's Categorization Output

Fill in the summary that AI produces (or your own from reviewing the output):

Category	Count	% of Total	Dominant Sentiment
Product Quality			
Delivery / Shipping			
Customer Service			
Technical / App			
Pricing / Value			
Other			
TOTAL	15	100%	

PART C — Escalation Prompt

Now write a follow-up prompt that asks AI to: (1) Identify the top 3 urgent issues requiring immediate action. (2) Draft a 3-sentence management alert for each. (3) Recommend 3 concrete process improvements.

Your escalation & recommendations prompt:

Record AI's top 3 recommended improvements:

PART D — Developer Extension

■■■ Automated Feedback Classifier

Ask AI to write a Python script that reads feedback.csv, sends each row's text to the OpenAI API (model: gpt-4o-mini) with a classification prompt, and writes the result to feedback_classified.csv. Include: batch processing (10 rows per call), basic error handling, and a final print of sentiment breakdown percentages.

■■■ Automated classifier script prompt:



5

Data-to-Report: Full Chain

■ General + ■■ Developer ● Advanced ■ 20 minutes

OBJECTIVE

Chain a complete 5-turn AI workflow — from raw data to finished executive report — using the sales dataset. Practice multi-turn workflow design with a defined output at each stage.

PART A — Design the 5-Turn Chain

Before writing prompts, design each turn in the table below. Define what goes *in* and what should come *out*:

Turn	Purpose	Input to AI	Expected Output
1			
2			
3			
4			
5 — Report	Generate final executive report	Insights from Turn 4	Formatted report, max 300 words, formal tone

PART B — Write the Turn 5 Report Prompt

Using the Report Generator Template from the Explanation Guide, write the full Turn 5 prompt that produces the executive management report from the sales analysis:

Turn 5 — Executive Report prompt (full CRTFC):

PART C — Report Quality Review

After receiving the report from AI, evaluate it against these criteria:

Quality Criterion	Met? (✓/X/Partial)	Comments / What to fix
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Leads with key finding		
All figures are accurate		
Within word limit		
Formal professional tone		
Includes actionable recommendations		
No invented data		

SECTION 3 | Building AI Mini-Automations

6

Build Your Sales Summary Automation

■ General + ■■ Developer ● Intermediate ■ 30 minutes

OBJECTIVE

Design and test a reusable Sales Summary Automation template. You will build the complete [PLACEHOLDER]-based prompt, test it with the sample data, and evaluate the output quality.

PART A — Build the Template

Using the Sales Summary Automation Template from the Explanation Guide as your starting point, adapt it for your own work context. Create a reusable template with your actual department, region names, and relevant metrics. Fill in every [PLACEHOLDER]:
Data: [PASTE SALES DATA]

Element	Template Text with [PLACEHOLDERS]
Prompt	Role: Sales Analyst. Task: Create a weekly sales summary report. Format: Max 300 words, professional tone. Constraints: Do not invent data.
Role	Act as a _____
Task	Write a _____ for the period: _____
Data	[PASTE DATA HERE]
Section 1	HEADLINE: _____
Section 2	KEY METRICS TABLE: _____
Section 3	TOP PERFORMERS: _____
Section 4	ISSUES / RISKS: _____
Section 5	RECOMMENDATIONS: _____
Format	Max _____ words. Tone: _____
Constraints	Do not: _____

PART B — Test Your Template

Run your completed template with the sales dataset. Then answer the following evaluation questions:

Question	Your Answer
Did the output follow the exact structure you defined?	
Were all figures accurate relative to the data?	

Was the tone appropriate for the intended audience?	
What would you change in the template for next time?	
Estimated time saved vs writing this manually:	

PART C — Developer Extension

■■■ Make Your Template Production-Ready

Ask AI to: (1) Convert your prompt template into a Python function called `generate_sales_report(csv_path, week_date, region)` that reads a CSV, calls the OpenAI API with your template, and returns the formatted report as a string. (2) Add a docstring explaining parameters and expected output. (3) Write one unit test that mocks the API call and validates the output structure.

■■■ Python template function request prompt:

7

Build a Task Planning Workflow

■ General + ■■ Developer ● Advanced ■ 30 minutes

OBJECTIVE

Design a complete AI task planning automation using a real or realistic project scenario. You will write the prompt, run it, and produce a structured prioritized task plan.

PART A — Define Your Scenario

Choose one of the scenarios below or describe your own real project context:

Scenario	Description	Select ✓
A — Office Relocation	Your team of 12 must move offices in 6 weeks. Tasks include IT setup, supplier notifications, HR updates, and communication plan.	
B — Product Launch	Launch a new service in 30 days: marketing brief, website update, sales training, customer communication, and monitoring plan.	
C — System Migration	Migrate from old CRM to new platform: data export, mapping, testing, staff training, and go-live cutover plan.	
D — Your Own Project	Describe your own real upcoming project in the box below.	

If Scenario D — describe your project context:

PART B — Write the Task Planning Prompt

Using the Task Planning Automation Template, write your complete prompt for the selected scenario:

Your task planning prompt (full CRTFC with your scenario):

PART C — Evaluate the Output

Record the key outputs from AI's task plan:

Output Element	AI's Output (summarized)	Quality (1–5)
Total tasks identified		
High priority tasks (count)		
Critical path identified?		
Top 3 risks listed		
Sprint allocation logical?		
Missing info flagged?		

PART D — Developer Extension

■■■ Engineering Sprint Planning

Use the developer Task Planning template to plan a 2-week engineering sprint. Provide 8–10 realistic Jira-style tickets (e.g. 'Implement JWT authentication endpoint', 'Write unit tests for payment service', 'Fix pagination bug on dashboard'). Ask AI to: estimate story points (Fibonacci), identify dependencies, flag unclear tickets, and allocate into a 40-point sprint.

■■■ Engineering sprint planning prompt:

SECTION 4 | No-Code Automation Design

8 Design an AI-Powered Automation Workflow

■ General + ■■ Developer ● Advanced ■ 30 minutes

OBJECTIVE

Design a conceptual automation workflow using one of the no-code tools covered in Day 3 (Make, Zapier, n8n, or Power Automate). You will map the trigger-to-output chain, identify the AI step, and write the prompt template for that step.

PART A — Select Your Workflow Pattern

Choose the workflow pattern you will design, or define your own:
Workflow: Weekly Report Automation
Trigger: New weekly sales data.

Pattern	Description	Select ✓
AI Step: Analyze data and create report. Output: Management report. Email Triage Destination: Email. Tool: n8n.	Auto-classify incoming emails by urgency, draft reply, send to folder	
Weekly Report Analyze [SALES_DATA], summarize performance, identify risks, and provide recommendations.	Scheduled: pull data from sheet → AI writes report → email to team	
Feedback Router	Form submission → AI classifies sentiment → route to Airtable + Slack alert	
Meeting Brief	Calendar event trigger → AI summarizes context → prep brief to attendees	
Custom / Your Own	Describe a workflow from your own work context	

PART B — Map the Full Automation Chain

Complete the trigger-to-output chain for your selected workflow:

Chain Step	Your Workflow Detail
TRIGGER What starts this workflow?	
APP / SOURCE Where does the trigger come from?	
FILTER (optional) Any condition that must be met?	
AI STEP What does AI do at this stage?	

AI TOOL Which model / service?	
OUTPUT What is produced?	
DESTINATION APP Where does output go?	
NOTIFICATION Who is alerted?	
AUTOMATION TOOL Make / Zapier / n8n / Power Automate	

PART C — Write the AI Step Prompt Template

Write the full CRTFC prompt template that will run inside your AI step. Include [PLACEHOLDERS] for the variable inputs that change each time the workflow runs:

AI Step prompt template (with [PLACEHOLDERS]):

PART D — Developer Extension

■■■ n8n Workflow JSON Design

Ask AI to generate a conceptual n8n workflow JSON structure for your chosen pattern. The JSON should define: trigger node type, HTTP request node (for the AI API call), a basic IF node for filtering, and a final action node. Ask AI to also explain which n8n node handles each step and what credentials are needed.

■■■ n8n workflow design prompt:

SECTION 5 | Capstone Workflow Lab

The Capstone Lab is your culminating deliverable for the full 3-day programme. Tasks 9 and 10 together form your Capstone submission: Task 9 is the **Design & Build** phase, Task 10 is the **Presentation Preparation** phase.

9

Capstone — Design & Build Your Workflow

■ General + ■■ Developer ● Capstone ■ 25 minutes

OBJECTIVE

Design and build a complete, end-to-end AI-powered workflow that solves a real problem from your own work context. This is your personal capstone — it should reflect something you will actually use after this programme.

CAPSTONE DESIGN CANVAS

Workflow Name: AI Weekly Sales Report Assistant
Complete Every field. This canvas is the foundation of your Task 10 presentation.

Problem: Manual weekly reports take time. Current State: Spreadsheet calculations and manual emails.	
Trigger: New sales file.	
PROBLEM STATEMENT (What takes too long, is [DATA] invalid, or quality is poor?)	
Output: Executive report.	
Automation: n8n.	
Review: Sales manager checks accuracy.	
Time Saved: 1-1.5 hours/week	
CURRENT STATE (How is this done today? Time cost?)	
TRIGGER (What event starts the workflow?)	
INPUT DATA (What goes in? Format? Source?)	
AI PROMPT TEMPLATE (Write your full CRTFC prompt with [PLACEHOLDERS])	
OUTPUT (Format, word/page limit, audience, destination)	
AUTOMATION TOOL (Make / Zapier / n8n / Power Automate / Manual trigger)	
REVIEW STEP (Who reviews? How long? What do they check?)	
TIME SAVED (Estimated hrs/week or hrs/month)	
RESPONSIBLE AI CHECK (Any privacy/accuracy concerns? How are they handled?)	

LIVE BUILD — Test Your Workflow Now

Run your workflow live during the session. Paste your prompt into your AI tool with real (anonymized) or realistic sample input data. Record the result:

Paste or summarize your AI output here:

■ Quality Self-Check Before Task 10

Check all that apply to your workflow:

- The problem is real and recurring — not a one-off task
- The trigger is clearly defined
- The AI step has a full CRTFC prompt template with [PLACEHOLDERS]
- The output format, audience, and destination are specified
- I have tested the prompt and reviewed the output
- I have considered privacy — no real PII in the prompt
- I have estimated the time saved

10

Capstone — Presentation & Peer Feedback

■ General + ■■ Developer ● Capstone ■ 20 minutes

OBJECTIVE

Prepare and deliver your 5-minute Capstone presentation. You will also give structured feedback to one peer presentation.

PART A — Presentation Slide Planner

Slide 1 Problem: Manual reporting challenge.
Slide 2 Workflow: Sales data ? non ? AP ? Report.
Slide 3 Prompt: CRTFC template.
Slide 4 Demo: Generated sales report.
Slide 5 Impact: Faster reporting and what you will say

		What you will say (key points — max 30 words per slide)	Time
1	The Problem		~60s
2	The Workflow		~90s
3	The Prompt	Show your CRTFC prompt template with [PLACEHOLDERS]	~60s
4	Live Demo / Output		~60s
5	Impact		~30s

PART B — Peer Feedback Form

After watching a peer's presentation, complete this structured feedback form. Feedback should be specific, constructive, and actionable.

Presenter name: _____
Workflow they presented: _____

Feedback Dimension	Rating (1–5)	Specific Comment
Problem was clearly defined		
Workflow logic was sound		

Prompt template was practical & reusable		
Output quality was demonstrated		
Time saved estimate was credible		
One thing I would improve:	—	
One thing I will steal for my own work:	—	

■ My Personal AI Playbook — Day 3 Additions

Add the templates and workflows you built today to your Personal AI Playbook. This becomes your reference document for using AI at work every day.

Task / Workflow	AI Tool	Template / Pattern	How Often	Est. Time Saved
Data upload & quality check	ChatGPT-4o	Upload Context + QA Prompt	Per dataset	30–60 min
Data cleaning	ChatGPT-4o	Cleaning Prompt Chain	Per dataset	1–2 hrs
Insight extraction + report	ChatGPT-4o / Claude	5-Turn Chain Template	Weekly	2–3 hrs/week
Feedback categorization	Claude	Feedback Categorization Template	Monthly	2–4 hrs/month
Sales summary automation	ChatGPT	Sales Summary Template	Weekly	90 min/week
Task planning	Claude	Task Planning Template	Per project	1 hr/project
[Your Capstone Workflow]				
[Your Custom Addition]				

■ Programme Completion Checklist — 3-Day Summary

Day	Skill	Completed ✓
Day 1	Applied CRTFC framework to write structured prompts	
Day 1	Used advanced techniques: Chain of Thought, Few-Shot	
Day 1	Built a reusable prompt template with [PLACEHOLDERS]	
Day 1	Designed a personal AI workflow using the IMTR cycle	
Day 2	Designed a multi-step AI workflow (Input → Process → Output)	
Day 2	Used role-based and constraint prompting for professional output	
Day 2	Produced an executive memo, meeting minutes, or SOP	
Day 2	Applied multi-document analysis and report-to-presentation	
Day 3	Uploaded data and ran the quality assessment workflow	
Day 3	Cleaned a dataset and verified AI's cleaning actions	
Day 3	Completed the 6-step Insight Extraction workflow	
Day 3	Built at least one mini-automation with a reusable template	
Day 3	Designed an automation chain using a no-code tool	
Day 3	Completed and presented the Capstone Workflow Lab	
All Days	Built a Personal AI Playbook with reusable templates	

■ Congratulations on completing the 3-Day AI for Office Automation programme. Your next step is to run your Capstone workflow on real data in the first week back at work.